Here are the solutions to the questions presented in the images. Since these are for a 15-mark evaluation, the answers are structured with detailed definitions, theoretical explanations, step-by-step calculations, and interpretations.

Q1. Central Tendency & Calculation

Part A: Theory (What is Central Tendency?)

Definition:

Central tendency (or measure of central tendency) is a statistical measure that identifies a single value as representative of an entire distribution. It aims to provide an accurate description of the entire data using one number, often called the "average."

Types of Central Tendency:

1. Mean (Arithmetic Mean): The sum of all observations divided by the total number of observations. It is the most common average.

2. Median: The middle value of a dataset when it is arranged in ascending or descending order. It divides the distribution into two equal halves.

3. Mode: The value that appears most frequently in the dataset.

4. Geometric Mean (GM): Used for growth rates; the nth root of the product of n numbers.

5. Harmonic Mean (HM): Used for rates and averages of speeds; the reciprocal of the arithmetic mean of the reciprocals.

Part B: Calculation

Data Analysis:

The data provided is a Discrete Series (Specific values of X with corresponding Frequencies F).

Note: The frequencies are strictly increasing. While this often suggests "Cumulative Frequency," the column is labeled 'F'. We will treat 'F' as the frequency for this solution.

Table for Calculation:

X (Variable) | F (Frequency)

10 | 15

20 | 35

30 | 60

40 | 84

50 | 96

60 | 127

70 | 198

80 | 250

1. Mean (X■):

2. Median (M):

For discrete series, Median is the value of the N/2 th item.

Looking at the CF column, the value 433 falls within the cumulative frequency of 615.

The X value corresponding to CF 615 is 70.

3. First Quartile (Q1):

Q1 is the value of the N/4 th item.

Looking at the CF column, 216.5 is greater than 194 but less than 290.

So it falls in the group with CF 290.

The X value corresponding to CF 290 is 50.

4. Third Quartile (Q3):

Q3 is the value of the 3N/4 th item.

Looking at the CF column, 649.5 falls in the group with CF 865.

The X value corresponding to CF 865 is 80.

Q2. Skewness and Kurtosis

Part A: Definitions

Skewness:

Skewness measures the asymmetry of the probability distribution of a real-valued random variable about its mean.

Symmetrical: Skewness = 0

Positive Skew: Tail on the right side is longer

Negative Skew: Tail on the left side is longer

Kurtosis:

Kurtosis is a measure of the "tailedness" of the probability distribution. It describes the peak of the curve.

Mesokurtic: Normal distribution (Kurtosis = 3)

Leptokurtic: High peak, fat tails

Platykurtic: Flat peak, thin tails

Part B: Test from Moments

Given:

First four central moments.

Testing Skewness:

The coefficient of skewness based on moments.

Interpretation:

Since the value is positive and close to zero, the distribution is Positively Skewed and nearly symmetrical.

Testing Kurtosis:

The coefficient of kurtosis based on moments.

Interpretation:

Since the value is approximately 3, the distribution is Mesokurtic.

Q3. Regression Analysis

Definition:

Regression analysis is a statistical method used to estimate the strength and direction of the relationship between two or more variables. It is used to predict the value of a dependent variable (Y) based on the known value of an independent variable (X).

Lines of Regression:

1. Line of Y on X:

Used to predict Y when X is given.

Equation: $Y = a + bX$

2. Line of X on Y:

Used to predict X when Y is given.

Equation: $X = a + bY$

Method of Least Squares:

The Least Squares method minimizes the sum of the squares of deviations between observed and estimated values.

Normal Equations:

$\Sigma Y = na + b\Sigma X$

$\Sigma XY = a\Sigma X + b\Sigma X^2$

Q4. Correlation & Calculation

Part A: Definitions

Correlation:

Correlation is a statistical technique that shows the degree of relationship between two variables. The value of correlation ranges from -1 to +1.

Scatter Diagram:

A graphical method to study correlation by plotting data points on X and Y axes.

Part B: Calculation

Pearson’s Correlation Coefficient Formula:

$r = \Sigma xy / \sqrt{(\Sigma x^2 \Sigma y^2)}$

Conclusion:

There is a high degree of positive correlation between X and Y. As X increases, Y also increases.